

Exact iid universality for unshifted binary embeddings: projective direction is the complete invariant

Candidate partial-result packet for arXiv:1512.04433

11 August 2026

Status. Candidate substantial partial result, likely valid, pending human review. The theorem resolves an exact iid-row formalization of the source question. It does not solve the broader dependent fast-transform or rate-comparison problem.

1 Source question and scope

Oymak and Recht study sign/Hamming embeddings of subsets of the unit sphere and ask which matrix ensembles can exhibit Gaussian behavior [1]. In the numerical-experiments discussion on PDF page 16, they write:

“Following from this example, it would be interesting to study the class of matrices that has the same empirical behavior as a Gaussian. For linear embedding this question has been studied extensively [9,19] and it remains as an open problem whether results of similar flavor would hold for binary embedding.”

$$\delta_{n-\text{lin}} = \left\| \mathbf{x}, \mathbf{y} \right\|_{\ell_2} \sup_{\mathbf{x}, \mathbf{y} \in K} \left| \left\| \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \right\|_{\ell_2} - \left\| \mathbf{x}, \mathbf{y} \right\|_{\ell_2} \right|.$$

Figure 1b shows the comparison of δ_{bin} and $\delta_{n-\text{lin}}$. We observe that linear embedding results in lower distortion but their behavior is highly similar. Figure 1 shows that linear and binary embedding performs on par and linear embedding does not provide a significant advantage. This is consistent with the main message of this work.

In Figure 2a we compare Gaussian embedding with fast binary embedding given by $\mathbf{A} = \mathbf{SDGD}^* \mathbf{R}$ as described in Section 6.3. We observe that Gaussian yields slightly better bounds however both techniques perform on par in all regimes. This further motivates theoretical understanding of fast binary embedding which significantly lags behind linear embedding. Sparse matrices are another strong alternative for fast multiplication and efficient embedding [8]. Figure 2b contrasts Gaussian embedding with sparse Gaussian where the entries are 0 with probability 2/3. Remarkably, the distortion dependence perfectly matches. Following from this example, it would be interesting to study the class of matrices that has the same empirical behavior as a Gaussian. For linear embedding this question has been studied extensively [9, 19] and it remains as an open problem whether results of similar flavor would hold for binary embedding.

Figure 1: The source passage, rendered from page 16 of arXiv:1512.04433.

The phrase “same empirical behavior” is intentionally broad. This packet isolates the exact iid, unshifted formulation: rows are independent copies of one random vector, the quantizer is the sign at threshold zero, and “same” means equality in law of the complete array of pairwise empirical Hamming distances on every finite set. In this formulation there is a sharp answer.

2 Definitions and statement

Fix $n \geq 1$. A random vector $A \in \mathbb{R}^n$ is called *hyperplane-nondegenerate* if

$$\mathbb{P}\{\langle A, x \rangle = 0\} = 0 \quad (x \in \mathbb{S}^{n-1}).$$

This condition in particular implies $A \neq 0$ almost surely. Its *projective direction* is the unoriented line $[A] \in \mathbb{RP}^{n-1}$. Let $\bar{\mu}$ denote the law of $[A]$, and let μ be its even lift to $\mathbb{S}^{n-1}$: conditionally on a line, choose either unit representative with probability $1/2$. Write σ for Haar probability on the sphere.

For iid rows $A_1, \dots, A_m$ and $x, y \in \mathbb{S}^{n-1}$ set

$$h_m(x, y) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{\text{sgn}\langle A_j, x \rangle \neq \text{sgn}\langle A_j, y \rangle\}, \quad d_{\text{ang}}(x, y) = \frac{\arccos\langle x, y \rangle}{\pi}.$$

The population separation metric is

$$d_\mu(x, y) = \mathbb{P}\{\text{sgn}\langle A, x \rangle \neq \text{sgn}\langle A, y \rangle\}.$$

Theorem 2.1 (Exact iid projective universality). *Let $A \in \mathbb{R}^n$ be hyperplane-nondegenerate. The following are equivalent.*

1. *The projective direction $[A]$ is Haar-uniform on $\mathbb{RP}^{n-1}$.*
2. *For every finite $K \subset \mathbb{S}^{n-1}$ and every $m \geq 1$, the joint array $(h_m(x, y))_{x, y \in K}$ has exactly the same law as for m iid standard Gaussian rows.*
3. *For every $x, y \in \mathbb{S}^{n-1}$,*

$$d_\mu(x, y) = d_{\text{ang}}(x, y).$$

Consequently, projective direction is the complete invariant for exact iid, unshifted Gaussian universality of pairwise binary distances.

Corollary 2.2 (Any non-Haar law has a two-point obstruction). *If $[A]$ is not projective Haar, there exist $x, y \in \mathbb{S}^{n-1}$ and $b > 0$ such that*

$$|d_\mu(x, y) - d_{\text{ang}}(x, y)| = b.$$

For iid copies of A and every $0 < \delta < b$,

$$\mathbb{P}\{|h_m(x, y) - d_{\text{ang}}(x, y)| \leq \delta\} \longrightarrow 0.$$

3 Proof intuition

Pairwise sign disagreements ignore both the row length and a simultaneous flip of all signs, so the row's projective direction is the only possible invariant. Haar projective direction is plainly sufficient. The substantive point is necessity.

Subtract Haar measure from the even spherical lift of the row law. Equality of all separation probabilities says that this signed measure annihilates every product of two hemisphere-sign functions. The hemisphere-sign transform has a nonzero Funk–Hecke eigenvalue in every odd harmonic degree, so every odd spherical harmonic can be recovered from such sign functions. The signed measure therefore annihilates every product of two odd harmonics. Those products span the even spherical polynomials, which determine an even measure. Hence the measure difference vanishes.

This gives exact classification in the iid model, but not the source paper's dependent fast-transform problem. It also reveals a sharp obstruction: ordinary isotropy and subgaussian tails do not control projective direction.

4 Proof of the characterization

For $x \in \mathbb{S}^{n-1}$ define the hemisphere-sign function

$$s_x(u) = \text{sgn}\langle u, x \rangle, \quad u \in \mathbb{S}^{n-1}.$$

Values on the equator are immaterial under the nondegeneracy assumption and under Haar measure.

Lemma 4.1 (Odd hemisphere transform is invertible by degree). *For every odd spherical harmonic Y on $\mathbb{S}^{n-1}$ there is a nonzero constant λ_Y such that*

$$\int_{\mathbb{S}^{n-1}} s_x(u) Y(x) \, d\sigma(x) = \lambda_Y Y(u).$$

The constant depends only on the harmonic degree (and on n).

Proof. The Funk–Hecke formula diagonalizes every zonal convolution on spherical harmonics; see, for example, [2, Chapter 1]. Applied to the kernel $t \mapsto \text{sgn } t$, it gives the asserted identity. It remains to verify that the eigenvalue is nonzero in every odd degree.

Suppose first that $n \geq 3$, put $\alpha = (n-2)/2$, and let the odd degree be $\ell = 2k+1$. Up to nonzero normalization factors, its eigenvalue is

$$\begin{aligned} I_\ell &= \int_{-1}^1 \text{sgn}(t) C_\ell^\alpha(t) (1-t^2)^{\alpha-1/2} \, dt \\ &= 2 \int_0^1 C_\ell^\alpha(t) (1-t^2)^{\alpha-1/2} \, dt, \end{aligned}$$

where C_ℓ^α is the Gegenbauer polynomial. Rodrigues' formula says that the integrand is a nonzero constant times

$$\frac{d^\ell}{dt^\ell} (1-t^2)^{\ell+\alpha-1/2}.$$

After integration, the derivative of order $\ell-1$ vanishes at $t=1$, because the remaining order of vanishing is $\alpha+1/2 > 0$. At $t=0$ it is

$$(2k)!(-1)^k \binom{\ell+\alpha-1/2}{k},$$

which is nonzero. Thus $I_\ell \neq 0$.

For $n=2$, write the circle variable as an angle. The Fourier series of $\text{sgn}(\cos \theta)$ has a nonzero coefficient at every odd frequency, so the same conclusion holds for every odd circular harmonic. When $n=1$ the projective space is a singleton and no inversion is needed. $\square$

Lemma 4.2 (Separation metric determines an even measure). *Let μ be an even probability measure on $\mathbb{S}^{n-1}$ that gives every great subsphere measure zero. If*

$$\int \mathbf{1}_{\{s_x(u) \neq s_y(u)\}} \, d\mu(u) = d_{\text{ang}}(x, y) \quad (x, y \in \mathbb{S}^{n-1}),$$

then $\mu = \sigma$.

Proof. If $n = 1$, then $\mathbb{R}P^0$ is a singleton and the conclusion is immediate. Assume henceforth that $n \geq 2$.

Let $\nu = \mu - \sigma$. It is an even finite signed measure of total mass zero. Rotational invariance in the plane spanned by x and y gives the classical random-hyperplane identity

$$\int \mathbf{1}_{\{s_x \neq s_y\}} d\sigma = \frac{\arccos\langle x, y \rangle}{\pi}.$$

Since $\mathbf{1}_{\{s_x \neq s_y\}} = (1 - s_x s_y)/2$ away from the two equators, the hypothesis is equivalent to

$$\int_{\mathbb{S}^{n-1}} s_x(u) s_y(u) d\nu(u) = 0 \quad (x, y \in \mathbb{S}^{n-1}). \quad (1)$$

Let Y and Z be arbitrary odd spherical harmonics. By Lemma 4.1, each is a nonzero scalar multiple of its hemisphere-sign integral representation. Fubini's theorem and (1) therefore give

$$\int_{\mathbb{S}^{n-1}} Y(u) Z(u) d\nu(u) = 0.$$

These products span all even spherical polynomials. Indeed, if P is the restriction to the sphere of an even polynomial, then on the sphere

$$P(u) = \sum_{i=1}^n u_i (u_i P(u)).$$

Each u_i is an odd degree-one harmonic, while the odd spherical polynomial $u_i P$ is a finite sum of spherical harmonics by the standard harmonic decomposition of polynomials. Hence ν annihilates P .

Spherical polynomials are uniformly dense in continuous functions on the sphere, and symmetrization makes the even ones dense in the continuous even functions. Thus ν annihilates every continuous even function. Since ν itself is even, it also annihilates the odd part of every continuous function. The Riesz representation theorem yields $\nu = 0$, proving $\mu = \sigma$. $\square$

Proof of Theorem 2.1. Suppose (1) holds. A pairwise disagreement pattern induced by one row is unchanged when that row is rescaled or negated, hence it is a measurable function only of its projective direction. A standard Gaussian row also has projective Haar direction. The per-row joint cut patterns on any finite K therefore have the same law; taking m iid products and averaging proves (2).

Condition (2), applied to $K = \{x, y\}$ (already with $m = 1$), gives equality of the expected disagreement indicators. The Gaussian random-hyperplane identity gives (3).

Finally, if (3) holds, apply Lemma 4.2 to the even lift μ . It follows that $\mu = \sigma$, which is equivalent to projective Haar uniformity. Thus (3) implies (1). $\square$

Proof of the corollary. The contrapositive of Theorem 2.1 gives x, y with nonzero bias b . For this fixed pair the summands defining $h_m(x, y)$ are iid Bernoulli variables of mean $d_\mu(x, y)$. The strong law gives $h_m(x, y) \rightarrow d_\mu(x, y)$ almost surely. Therefore the displayed probability tends to zero whenever $\delta < b$. $\square$

5 A sharp obstruction to linear-style universality

Corollary 5.1 (Isotropic subgaussian rows need not be binary-universal). *Let $n \geq 2$ and let $A = (\varepsilon_1, \dots, \varepsilon_n)$ have independent Rademacher coordinates. The row is isotropic and bounded subgaussian. Yet for every $0 < t < 1$, with*

$$x = e_1, \quad y = \frac{e_1 + te_2}{\sqrt{1+t^2}},$$

one has, for every row realization,

$$\operatorname{sgn}\langle A, x \rangle = \operatorname{sgn}\langle A, y \rangle.$$

Consequently $h_m(x, y) = 0$ deterministically for every m , while

$$d_{\text{ang}}(x, y) = \frac{\arctan t}{\pi} > 0.$$

Thus no distortion smaller than $\arctan(t)/\pi$ is possible for this two-point set, regardless of the number of rows.

Proof. Isotropy is $\mathbb{E}AA^T = I_n$, and the standard Rademacher moment-generating function bound gives a dimension-free subgaussian constant. Moreover

$$\operatorname{sgn}\langle A, x \rangle = \varepsilon_1, \quad \operatorname{sgn}\langle A, y \rangle = \operatorname{sgn}(\varepsilon_1 + t\varepsilon_2) = \varepsilon_1,$$

because $0 < t < 1$. The angle identity follows from $\langle x, y \rangle = (1 + t^2)^{-1/2}$. □

Remark 5.2. *The Rademacher law has atoms on some great subspheres, so it is intentionally presented as a direct obstruction to broad isotropic/subgaussian universality, not as an application of the nondegenerate equivalence theorem. The displayed pair itself never encounters a zero inner product.*

6 Verification, novelty boundary, and limitations

The proof was audited separately in `verification.md`. The reusable script

`code/check_gegenbauer_coefficients.py`

symbolically checks the odd coefficient integral for dimensions 3 through 8 and odd degrees 1 through 9, and exhaustively checks the two-dimensional Rademacher sign calculation for three rational values of t . These finite checks passed, but they are not proof; Lemma 4.1 and Corollary 5.1 give the general arguments.

A bounded novelty search checked the run’s registry, solution, attempt, and proof-gap indexes, as well as arXiv/web searches around binary embedding, projective direction, random hyperplanes, universality, and Funk–Hecke. Later work includes fast Gaussian circulant embeddings [3], structured quantized embeddings [4], and non-Gaussian tessellations using affine random hyperplanes (random shifts/dithering) [5]. No exact version of Theorem 2.1 was located. This is bounded-search evidence only, not a guarantee of novelty.

The result is partial relative to the source question in three ways.

- It assumes iid rows and therefore does not analyze dependent fast maps such as the structured transform highlighted by Oymak and Recht.

- It characterizes exact equality in law, not equality of sample complexity or distortion rates up to constants.
- It concerns the unshifted sign quantizer. Dithering can restore useful non-Gaussian results by changing the geometry of the hyperplanes.

Human-review recommendation. Review as a substantial partial result. The critical checks are the nonzero odd Funk–Hecke coefficient and the span of even spherical polynomials by products of odd harmonics.

References

- [1] S. Oymak and B. Recht, *Near-Optimal Bounds for Binary Embeddings of Arbitrary Sets*, arXiv:1512.04433 (2015).
- [2] F. Dai and Y. Xu, *Approximation Theory and Harmonic Analysis on Spheres and Balls*, Springer Monographs in Mathematics, Springer, 2013.
- [3] S. Dirksen and A. Stollenwerk, *Fast Binary Embeddings with Gaussian Circulant Matrices: Improved Bounds*, arXiv:1608.06498 (2016).
- [4] D. Huynh and R. Saab, *Fast Binary Embeddings, and Quantized Compressed Sensing with Structured Matrices*, arXiv:1801.08639 (2018).
- [5] S. Dirksen and S. Mendelson, *Non-Gaussian Hyperplane Tessellations and Robust One-Bit Compressed Sensing*, arXiv:1805.09409 (2018).